

R Laboratory - Session 2

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Advanced Statistics for Physics Analysis (2025-2026)

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Additional comments explaining the code in a more detailed way are not reported here, but are available on the corresponding .R source code

We are given a dataset of integer-valued observations, $D = \{x_1, x_2, \dots, x_N\}$ contained in `assignment2_changepoint_counts.csv`. The data represents the number of events observed in identical consecutive time intervals, where the order of the observations matters.

(1) Bayesian change-point model

We assume that the data D is generated from a Poisson model with a one unknown change point m , with respectively λ_1 and λ_2 the event rates before and after m . The parameters have Gamma priors $\lambda_1, \lambda_2 \sim \text{Gamma}(\alpha, \beta)$ and $m \sim \text{Unif}(\mathcal{M})$ where $\mathcal{M} = \{n_{\min}, n_{\min} + 1, \dots, N - n_{\min}\}$.

(1.a)

The likelihood of the data $p(D|\lambda_1, \lambda_2, m)$ is the product of the individual Poisson probabilities, separated at m :

$$p(D|\lambda_1, \lambda_2, m) = \prod_{i=1}^m p(x_i|\lambda_1) \prod_{i=m+1}^N p(x_i|\lambda_2)$$

Replacing with the Poisson distribution formula, we now have :

$$p(D|\lambda_1, \lambda_2, m) = \prod_{i=1}^m \left(\frac{\lambda_1^{x_i} e^{-\lambda_1}}{x_i!} \right) \prod_{i=m+1}^N \left(\frac{\lambda_2^{x_i} e^{-\lambda_2}}{x_i!} \right)$$

And finally we can use the cumulative sums defined in the next questions, $S_m = \sum_{i=1}^m x_i$ and $S_N = \sum_{i=1}^N x_i$, to simplify the expression and we have :

$$\Rightarrow p(D|\lambda_1, \lambda_2, m) = \frac{\lambda_1^{S_m} e^{-m\lambda_1} \lambda_2^{S_N - S_m} e^{-(N-m)\lambda_2}}{\prod_{i=1}^N x_i!}$$

(1.b)

In order to find the joint posterior distribution up to a proportionality constant we can apply Bayes theorem :

$$p(\lambda_1, \lambda_2, m|D) \propto p(D|\lambda_1, \lambda_2, m)p(\lambda_1)p(\lambda_2)p(m)$$

We know that the prior for m is a constant $\frac{1}{|\mathcal{M}|}$, and the Gamma priors for λ_1 and λ_2 are proportional to $\lambda^{\alpha-1}e^{-\beta\lambda}$. The precision up to a proportionality constant allows us to remove

the denominator of the likelihood ($\prod x_i!$, depends only on the data) and constant terms in the priors. We then have :

$$p(\lambda_1, \lambda_2, m|D) \propto \left(\lambda_1^{S_m} e^{-m\lambda_1} \lambda_2^{S_N - S_m} e^{-(N-m)\lambda_2} \right) \left(\lambda_1^{\alpha-1} e^{-\beta\lambda_1} \right) \left(\lambda_2^{\alpha-1} e^{-\beta\lambda_2} \right)$$

Which is equivalent to (after rearranging the terms) :

$$\implies \boxed{p(\lambda_1, \lambda_2, m|D) \propto \lambda_1^{S_m + \alpha - 1} e^{-(\beta+m)\lambda_1} \lambda_2^{S_N - S_m + \alpha - 1} e^{-(\beta+N-m)\lambda_2}}$$

(1.c)

To find the conditional posterior distribution for λ_1 given m and the data, we can isolate the terms in the joint posterior that depend only on λ_1 :

$$p(\lambda_1|\lambda_2, m, D) \propto \lambda_1^{S_m + \alpha - 1} e^{-(\beta+m)\lambda_1}$$

We recognize this mathematical form as the kernel of a Gamma distribution. The same logic applies to λ_2 , isolating the terms depending on λ_2 :

$$p(\lambda_2|\lambda_1, m, D) \propto \lambda_2^{S_N - S_m + \alpha - 1} e^{-(\beta+N-m)\lambda_2}$$

Therefore, conditionally on m , the rates follow :

$$\begin{aligned} &\implies \boxed{\lambda_1|m, D \sim \text{Gamma}(S_m + \alpha, \beta + m)} \\ &\quad \text{and} \\ &\implies \boxed{\lambda_2|m, D \sim \text{Gamma}(S_N - S_m + \alpha, \beta + N - m)} \end{aligned}$$

(1.d)

In order to derive the conditional posterior of m , we must isolate the terms in the joint posterior that depend on m and then we consider λ_1 and λ_2 as constants :

$$\begin{aligned} p(m|\lambda_1, \lambda_2, D) &\propto \lambda_1^{S_m} e^{-m\lambda_1} \lambda_2^{-S_m} e^{m\lambda_2} \\ p(m|\lambda_1, \lambda_2, D) &\propto \left(\frac{\lambda_1}{\lambda_2} \right)^{S_m} e^{-m(\lambda_1 - \lambda_2)} \end{aligned}$$

To make it a proper probability distribution, we must normalize it over all possible values of $m \in \mathcal{M}$:

$$\implies \boxed{p(m|\lambda_1, \lambda_2, D) = \frac{\left(\frac{\lambda_1}{\lambda_2} \right)^{S_m} e^{-m(\lambda_1 - \lambda_2)}}{\sum_{k \in \mathcal{M}} \left(\frac{\lambda_1}{\lambda_2} \right)^{S_k} e^{-k(\lambda_1 - \lambda_2)}}$$

This is a discrete distribution over $\mathcal{M}$ because the prior probability of m restricts it exclusively to the finite set of integers $\mathcal{M} = \{n_{min}, \dots, N - n_{min}\}$. The probability mass function assigns a specific normalized weight to each integer m in this set.

(1.e)

This model is a natural case for Markov Chain Monte Carlo because all full conditional posterior distributions ($p(\lambda_1|\dots)$, $p(\lambda_2|\dots)$, and $p(m|\dots)$) are available analytically and are easy to sample from. However, the exact joint posterior $p(\lambda_1, \lambda_2, m|D)$ does not have a recognizable closed form, and evaluating the exact marginal distributions for λ_1 and λ_2 would require computing a heavy mixture over all possible values of m . By using MCMC (and more specifically Gibbs sampling designed for this type of cases), we can bypass this analytical bottleneck because we iteratively sample from the simple known conditionals to empirically reconstruct the complex joint distribution.

(2) Gibbs sampling

(2.a)

A Gibbs sampler iteratively draws samples from the full conditional posterior distributions derived in the previous questions. Let $S_k = \sum_{i=1}^k x_i$ be the k cumulative sums of our dataset. For an iteration t , the three update steps are :

1. **Update $\lambda_1^{(t)}$** : Draw from $\text{Gamma}(S_{m^{(t-1)}} + \alpha, m^{(t-1)} + \beta)$
2. **Update $\lambda_2^{(t)}$** : Draw from $\text{Gamma}(S_N - S_{m^{(t-1)}} + \alpha, N - m^{(t-1)} + \beta)$
3. **Update $m^{(t)}$** : The unnormalized log-probability for a given candidate $k \in \mathcal{M}$ is :

$$\log p(k|\dots) = S_k \log \left(\frac{\lambda_1^{(t)}}{\lambda_2^{(t)}} \right) - k(\lambda_1^{(t)} - \lambda_2^{(t)})$$

To avoid numerical underflow or overflow when computing the normalized probabilities $p(m = k|\dots)$, we apply the log-sum-exp trick. We find the maximum log-probability $\max_j(\log p_j)$ and compute the normalized probabilities as :

$$P(m = k|\lambda_1, \lambda_2, D) = \frac{\exp(\log p_k - \max_j(\log p_j))}{\sum_{j \in \mathcal{M}} \exp(\log p_j - \max_j(\log p_j))}$$

These steps are implemented in R :

```
1  # 1 : updating lambda_1 from its Gamma conditional
2  S_m <- S[m]
3  alpha_1 <- S_m + alpha
4  beta_1 <- m + beta
5  l1 <- rgamma(1, shape = alpha_1, rate = beta_1)
6
7  # 2 : updating lambda_2 from its Gamma conditional
8  alpha_2 <- S_N - S_m + alpha
9  beta_2 <- N - m + beta
10 l2 <- rgamma(1, shape = alpha_2, rate = beta_2)
11
12 # 3 : updating m from its discrete conditional, using the log-sum-exp trick
13 log_probs <- numeric(length(M))
14 for (j in 1:length(M))
15 {
16   k <- M[j]
17   S_k <- S[k]
18   # calculating the log-probability for a given k
19   log_probs[j] <- S_k * log(l1 / l2) - k * (l1 - l2)
20 }
21
22 # normalizing to prevent numerical underflow or overflow
23 max_log_prob <- max(log_probs)
24 probs <- exp(log_probs - max_log_prob)
25 probs <- probs / sum(probs)
26
27 # sampling the new change point m based on the normalized probabilities
28 m <- sample(M, size = 1, prob = probs)
```

(2.b)

We run three separate MCMC chains with different m , λ_1 and λ_2 initial values to explore of the parameter space and check the convergence. The trace plots shows an immediate convergence, with all the chains overlapping around the same target distributions.

```
1 n_iteration <- 5000
2 set.seed(2107530) # for reproducibility
3
4 chain1 <- gibbs_sampler(n_iteration, initial_m = 10, init_l1 = 1, init_l2 = 15,
5   x, alpha, beta)
6 chain2 <- gibbs_sampler(n_iteration, initial_m = 60, init_l1 = 10, init_l2 =
7   10, x, alpha, beta)
8 chain3 <- gibbs_sampler(n_iteration, initial_m = 110, init_l1 = 15, init_l2 =
9   1, x, alpha, beta)
```

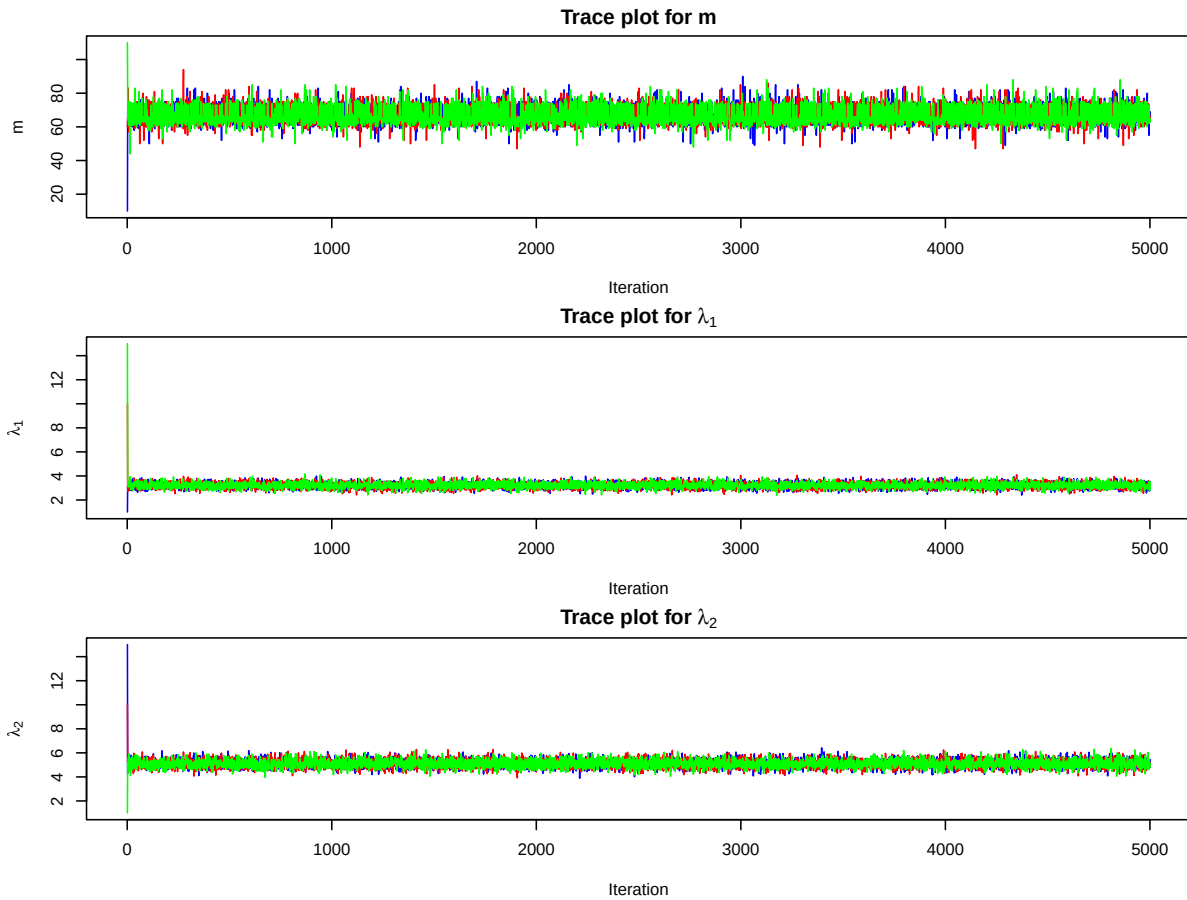


Figure 1: Trace plots for m , λ_1 , and λ_2 across the 3 chains (blue, red and green), including the initial values before the first iteration. The chains mix well and immediately converge to their stationary distributions.

(2.c)

By visually inspecting the trace plots, we observe that the chains leave their initialized states and reach the high-probability region almost instantaneously. To be highly and safely conservative and be sure to remove any residual influence of the starting values, an initial burn-in period of 1000 iterations was chosen for each chain. The remaining 4000 iterations of the three chains were combined into a final posterior pool of 12000 samples.

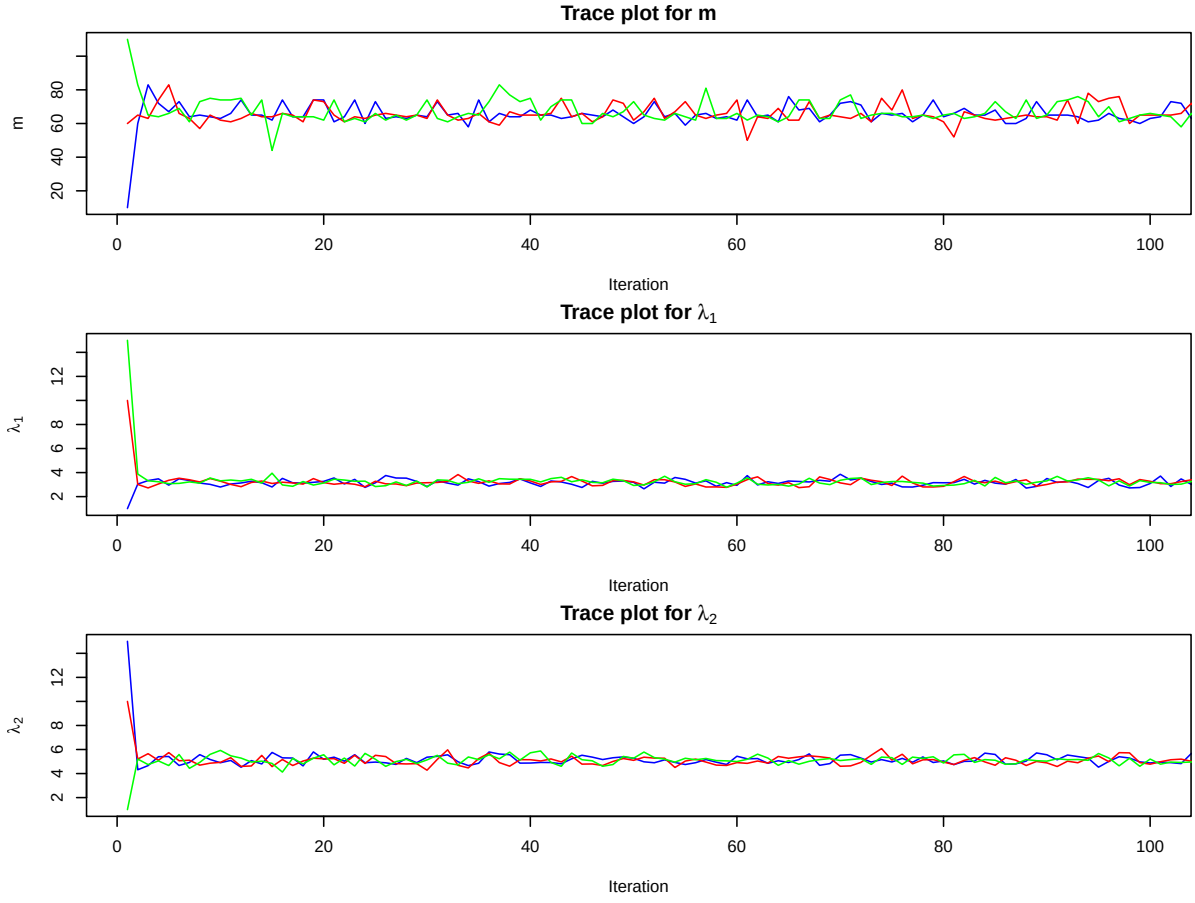


Figure 2: Trace plots for m , λ_1 , and λ_2 across the 3 chains (blue, red and green), showing only the first 100 iterations (including the initial values before the first iteration)

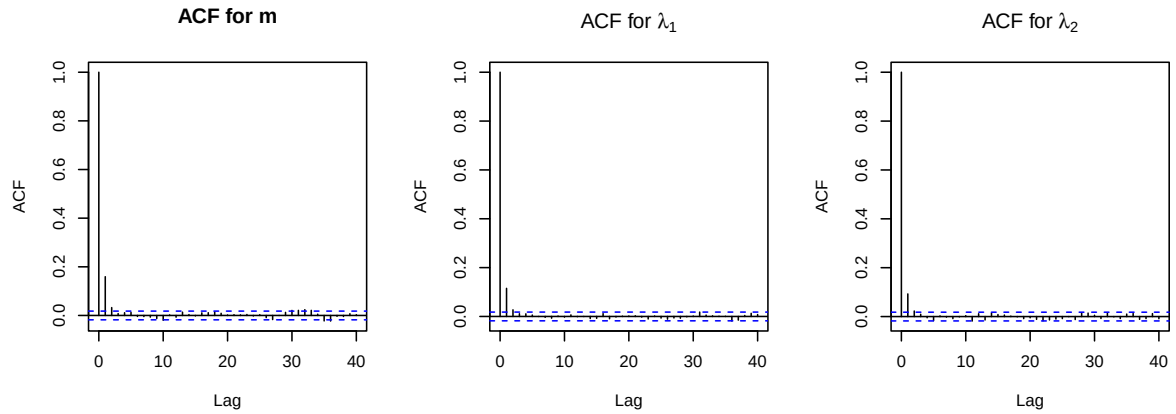


Figure 3: Autocorrelation function (ACF) for m , λ_1 , and λ_2 . The rapid drop to zero confirms the independence of the MCMC samples. The blue dashed lines (---) represent the 95% confidence bounds, approximately given by $\pm 1.96/\sqrt{N}$, where N is the sample size. Any lag exceeding these limits indicates a statistically significant autocorrelation.

(2.d)

The autocorrelation function (ACF) plots (Figure 3) show a very rapid drop to 0 for all three parameters, indicating excellent mixing properties. Using the `coda` package, we estimated the Effective Sample Sizes (ESS), which are roughly equal to the total number of samples, proving that the chains are producing highly independent draws :

Effective Sample Sizes (ESS) :

```
m          : 8700.091
lambda_1   : 9258.23
lambda_2   : 9683.319
```

(2.e)

By applying summary statistics to the combined posterior samples (and using a density estimator to extract the maximum a posteriori for the continuous variables), we obtain the results summarized in the table 1 :

Parameter	Mean	Median	Standard Deviation	Maximum A Posteriori	95% Credible Interval
m	66.28	65	5.12	64	[59, 77]
λ_1	3.21	3.20	0.23	3.19	[2.77, 3.68]
λ_2	5.11	5.10	0.33	5.12	[4.50, 5.77]

Table 1: Posterior summaries for the change-point model parameters

We also computed the correlation matrix of our posterior samples to investigate parameter dependencies, reported in the table 2 :

	m	λ_1	λ_2
m	1.000	0.313	0.248
λ_1	0.313	1.000	0.068
λ_2	0.248	0.068	1.000

Table 2: Correlation matrix of the combined posterior samples

We observe positive correlations between m and λ_1 (≈ 0.31) and between m and λ_2 (≈ 0.25). This is physically consistent : if the change point is shifted later into the higher-count regime, it absorbs higher values into the first interval (raising the estimated mean λ_1) while simultaneously concentrating the second interval strictly on the highest counts (raising the estimated mean λ_2).

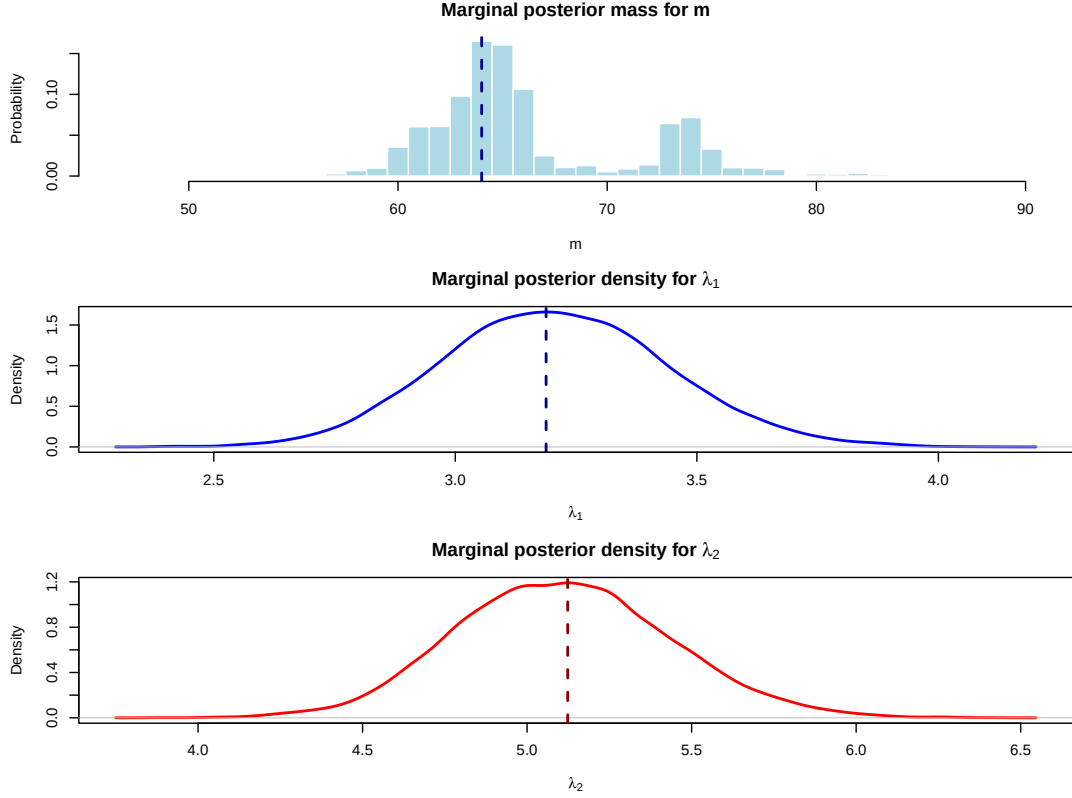


Figure 4: Marginal posterior distributions for m , λ_1 , and λ_2 . The dashed vertical lines represent the Maximum A Posteriori (MAP) estimates.

Figure 4 shows these marginal posterior distributions. We treat the continuous rates λ_1 and λ_2 using kernel density estimation, and the discrete change-point m is represented by its exact probability mass. The dashed lines indicate the respective MAP estimates, to visual confirm the strict separation between the pre-change and post-change rates.

(2.f)

We estimate the posterior probability $P(\lambda_2 > \lambda_1 | D)$ by computing the fraction of MCMC iterations where the drawn value of λ_2 is superior to λ_1 .

```
1 # counting the fraction of MCMC steps where l2 > l1
2 prob_l2_greater_l1 <- mean(posterior$l2 > posterior$l1)
3 cat('\n\nPosterior probability P(lambda_2 > lambda_1 | D) :',
4     prob_l2_greater_l1, '\n')
```

Posterior probability P(lambda_2 > lambda_1 | D) : 1

The probability that the event rate increased after the change point is 100%. This confirms that a definitive upward shift in the occurrence rate happened in the time series, specifically around the interval $m = 65$, where the average count jumped from roughly 3.2 to 5.1 events per interval.

(3) Posterior predictive distribution

(3.a)

The posterior predictive distribution for a new observation x_{new} is obtained by marginalizing the likelihood of the new observation over the joint posterior distribution of all the unknown parameters :

$$p(x_{new}|D) = \sum_{m \in \mathcal{M}} \iint p(x_{new}|\lambda_1, \lambda_2, m) p(\lambda_1, \lambda_2, m|D) d\lambda_1 d\lambda_2$$

x_{new} is an observation in the next time interval (at the time $N + 1$), so we are strictly evaluating a time where the index $i > m$, since the change point is heavily constrained by the prior to $m \leq N - n_{min}$. Therefore, the event rate has already transitioned, and the new observation depends uniquely on the second rate parameter λ_2 .

$$p(x_{new}|\lambda_1, \lambda_2, m) = p(x_{new}|\lambda_2) = Poisson(x_{new}|\lambda_2)$$

By integrating out the conditionally independent parameters (λ_1 and m), the equation simplifies to the expectation over the marginal posterior of λ_2 :

$$\Rightarrow \boxed{p(x_{new}|D) = \int Poisson(x_{new}|\lambda_2) p(\lambda_2|D) d\lambda_2}$$

(3.b)

From the law of total expectation, the posterior predictive mean $\mathbb{E}[x_{new}|D]$ corresponds to the posterior mean of λ_2 . We can estimate this empirically using our MCMC output by generating new observations $x_{new}^{(t)} \sim Poisson(\lambda_2^{(t)})$ for each sampled rate and computing their mean. Both the theoretical marginal mean and the simulated mean converge :

Posterior predictive mean $\mathbb{E}[x_{new} | D]$:

```
Theoretical (mean of lambda_2) : 5.108197
Simulated (mean of x_new)      : 5.11975
```

(3.c)

We want to compute the probability that the next observation exceeds twice the empirical average of our dataset, $\mathbb{P}(x_{new} > 2\bar{x}|D)$, where $\bar{x} = 4.05$. Using our simulated dataset `x_new_simu`, we find the proportion of elements strictly greater than $2\bar{x} = 8.1$.

```
Empirical average of data (x_bar) : 4.05
Threshold (2 * x_bar)             : 8.1
Probability P(x_new > 2 * x_bar | D): 0.08041667
```


(3.d)

To visualize the posterior predictive distribution, we made a histogram of the simulated values $x_{new}^{(t)}$. Breaks were offset by 0.5 to ensure the bars are correctly centered on integers.

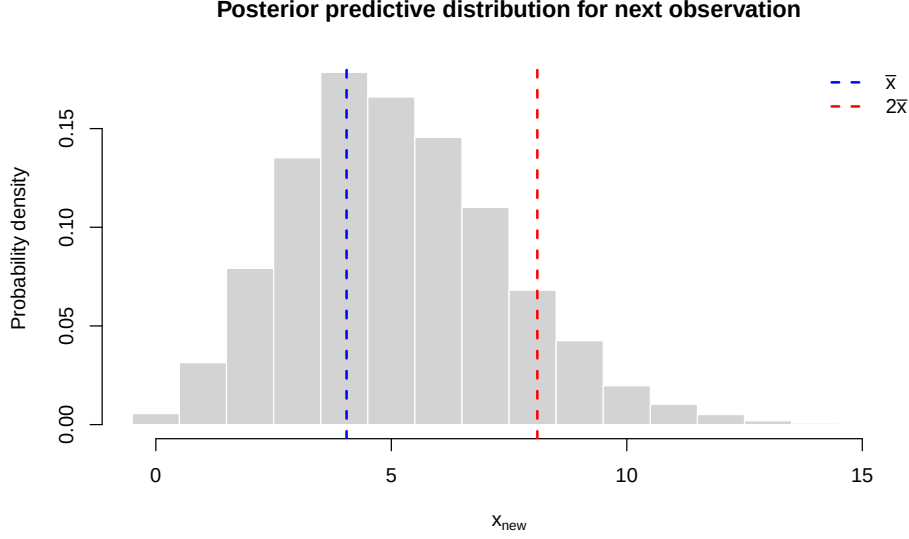


Figure 5: Posterior predictive distribution of the next observation x_{new} . The dashed blue line indicates the dataset empirical mean ($\bar{x} = 4.05$), and the red dashed line is the threshold limit ($2\bar{x} = 8.1$).

(4) Model comparison

We now compare the one-change-point model (M_1) with the simpler no-change model (M_0), where $x_i | \lambda \sim \text{Poisson}(\lambda)$ for all $i = 1, \dots, N$.

(4.a)

To find the marginal likelihood of the no-change model $p(D|M_0)$, we integrate the joint distribution over the single parameter λ :

$$p(D|M_0) = \int_0^\infty p(D|\lambda)p(\lambda) d\lambda$$

$$p(D|M_0) = \int_0^\infty \left(\prod_{i=1}^N \frac{\lambda^{x_i} e^{-\lambda}}{x_i!} \right) \left(\frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda} \right) d\lambda$$

We can factor out the constants and substitute $S_N = \sum_{i=1}^N x_i$:

$$p(D|M_0) = \frac{1}{\prod_{i=1}^N x_i!} \frac{\beta^\alpha}{\Gamma(\alpha)} \int_0^\infty \lambda^{S_N + \alpha - 1} e^{-(N+\beta)\lambda} d\lambda$$

The integral evaluates to the normalizing constant of a $\text{Gamma}(S_N + \alpha, N + \beta)$ distribution :

$$\Rightarrow p(D|M_0) = \frac{1}{\prod_{i=1}^N x_i!} \frac{\beta^\alpha}{\Gamma(\alpha)} \frac{\Gamma(S_N + \alpha)}{(N + \beta)^{S_N + \alpha}}$$

(4.b)

For the one-change-point model M_1 , we sum over the discrete uniform prior for m and integrate over the independent priors for λ_1 and λ_2 .

$$p(D|M_1) = \sum_{m \in \mathcal{M}} p(m) \iint p(D|\lambda_1, \lambda_2, m) p(\lambda_1) p(\lambda_2) d\lambda_1 d\lambda_2$$

Using our joint likelihood from (1.b) and separating the integrals for λ_1 and λ_2 :

$$p(D|M_1) = \frac{1}{|\mathcal{M}| \prod_{i=1}^N x_i!} \left(\frac{\beta^\alpha}{\Gamma(\alpha)} \right)^2 \sum_{m \in \mathcal{M}} \left[\int_0^\infty \lambda_1^{S_m + \alpha - 1} e^{-(m+\beta)\lambda_1} d\lambda_1 \right] \left[\int_0^\infty \lambda_2^{S_N - S_m + \alpha - 1} e^{-(N-m+\beta)\lambda_2} d\lambda_2 \right]$$

These are again Gamma integrals, evaluating them gives us the exact marginal likelihood :

$$\Rightarrow p(D|M_1) = \frac{1}{\prod_{i=1}^N x_i!} \frac{1}{|\mathcal{M}|} \left(\frac{\beta^\alpha}{\Gamma(\alpha)} \right)^2 \sum_{m \in \mathcal{M}} \frac{\Gamma(S_m + \alpha)}{(m + \beta)^{S_m + \alpha}} \frac{\Gamma(S_N - S_m + \alpha)}{(N - m + \beta)^{S_N - S_m + \alpha}}$$

(4.c)

The Bayes factor is defined as $BF_{10} = \frac{p(D|M_1)}{p(D|M_0)}$. The common factor in both models $\frac{1}{\prod x_i!}$ perfectly cancels out so we can omit it. To avoid numerical underflow, we computed the analytical formulas in log-space using R :

```
1 S <- cumsum(x)
2 S_N <- S[N] # cumulative sums
3
4 # Log marginal likelihood for M0 (no-change model)
5 log_ev_M0_core <- alpha * log(beta) - lgamma(alpha) +
6   lgamma(S_N + alpha) - (S_N + alpha) * log(N + beta)
7
8 # Log marginal likelihood for M1 (one-change-point model)
9 log_ev_M1_m_core <- numeric(length(M))
10
11 for (j in 1:length(M)) {
12   m_val <- M[j]
13   S_m <- S[m_val]
14
15   # Log evidence for a specific m
16   log_ev_M1_m_core[j] <- 2 * alpha * log(beta) - 2 * lgamma(alpha) +
17     lgamma(S_m + alpha) - (S_m + alpha) * log(m_val + beta)
18   +
19     lgamma(S_N - S_m + alpha) - (S_N - S_m + alpha) * log(
20     N - m_val + beta) }
21
22 log_prior_m <- -log(length(M)) # The prior probability for m is 1 / |M|
23
24 # To sum exp(log_ev_M1_m_core) without overflow, we use the log-sum-exp trick
25 max_log_ev <- max(log_ev_M1_m_core)
26 log_sum_exp_term <- max_log_ev + log(sum(exp(log_ev_M1_m_core - max_log_ev)))
27
28 # Final log marginal likelihood for M1
29 log_ev_M1_core <- log_prior_m + log_sum_exp_term
30
31 log_BF_10 <- log_ev_M1_core - log_ev_M0_core # computing log(BF)
32 BF_10 <- exp(log_BF_10)
```

Model Comparison (Log scale) :

Log Evidence M0 (without the common factor) : 190.2929

Log Evidence M1 (without the common factor) : 198.5858

Log Bayes Factor ($\ln BF_{10}$) : 8.29292

Bayes Factor (BF_{10}) : 4.00e+03

We can see that the log Bayes factor is strictly positive and very large ($\ln BF_{10} \approx 8.29$), and therefore the Bayes factor itself is $BF_{10} \approx 3995$. Following the standard Jeffreys scale, any $BF_{10} > 100$ (or $\ln BF_{10} > 4.6$) is considered a decisive evidence in favour of model M_1 . The model with a change point is massively favoured over the constant-rate model, confirming our MCMC findings that a significant structural increase in the event rate occurred during the observational period.

(5) Discussion

Based on the results from the previous sections, we can draw some conclusions :

- There is decisive evidence for a change point. The computed Bayes factor comparing the one-change-point model to the constant-rate model is extremely large ($BF_{10} \approx 3995$, or $\ln BF_{10} \approx 8.29$). According to the Jeffreys scale, this confirms that a structural change in the underlying data generation process occurred.
- The most plausible location for this structural shift is around the interval 64 or 65, which correspond respectively to the MAP estimate and the median of our posterior samples. The 95% credible interval strictly places the true change point between $m = 59$ and $m = 77$.
- The two rates are perfectly separated. The evaluated posterior probability $\mathbb{P}(\lambda_2 > \lambda_1 | D)$ is exactly 1. Moreover, their 95% credible intervals ($\lambda_1 \in [2.77, 3.68]$ and $\lambda_2 \in [4.50, 5.77]$) are entirely mutually exclusive. This clearly indicates a definitive upward shift in the event rate, jumping from roughly 3.2 to 5.1 events per interval.
- The three MCMC chains, initialized at widely dispersed starting values ($m = 10, 60, 110$), gave highly consistent results. The visual inspection of the trace plots showed that all chains instantly left their initialized states, converged immediately, and perfectly overlapped in the high-probability region.
- The autocorrelation functions dropped to nearly zero almost immediately after lag 1, indicating excellent mixing properties. Therefore the Effective Sample Sizes (ESS) are exceptionally high (ranging from ~ 8700 to ~ 9683 out of 12000 total valid iterations). This proves the generated samples are highly independent, making our posterior summaries and derived credible intervals highly reliable.
- Hamiltonian Monte Carlo (HMC) explores the parameter space using simulated physics (the Hamiltonian dynamics), which is a process that fundamentally requires computing the gradient of the log-posterior with respect to the target parameters. The change point m is a discrete parameter, so the posterior space is neither continuous nor differentiable with respect to m . Therefore, HMC cannot evaluate the necessary gradients and cannot be applied directly to the full joint state $(\lambda_1, \lambda_2, m)$ without mathematically marginalizing m out of the model first.